

Data sources

The first data source that we selected and adapted data from is the Canadian Anti-Fraud Centre (CAFC) dataset called “Canadian Anti-Fraud Centre Reporting Data” ([Canadian Anti-Fraud Centre Fraud Reporting System Dataset - Open Government Portal](#)). It was published in July 2023 and last updated in October 2025. It has the licence Open Government Licence - Canada. From this source, we randomly selected 200 reports, where Complaint Type = “Victim” and there was exactly one victim per report. The selected CAFC data is used to populate our Fraud_type, Communication_Channel, Fraud_Incident, Report and Organisation tables

The second data source that we have chosen is the Consumer Sentinel Network Data Book 2024 from the Federal Trade Commission (FTC) in the United States ([Consumer Sentinel Network Data Book 2024 | Federal Trade Commission](#)). It was published in March 2025. It has no licence, because work from the US government is in the public domain, so it is free to use. From this source we used the number of reports per fraud type for 2022, 2023 and 2024 (Appendix B3), and the reports and losses per contact method and per age group (pages 12 and 13). We only kept the fraud categories, which gave 123 report counts and 16 loss statistics.

CAFC and FTC are not subsets of each other.

Integrating data into the database

For the CAFC data, the schema had to be updated to accommodate the way age was notated. In the data, age is collected as a range as opposed to an exact number. So the attribute Age was removed and replaced with the attribute Age_range. In addition, Financial_loss_amount in the Fraud_Incident table now allows NULL values to facilitate the data.

Alongside this, the CAFC data also required some data cleaning and transformation. There were no empty cells, instead missing values were represented as text, such as “Not Available” and “Unknown” in the gender column, and “Not Available”, “Unknown” and “Deceased” in age. These missing values were replaced with NULL in Age_range and Gender. Besides that, the financial loss data also needed to be transformed. The dollar sign and comma separators, for example \$1,000.00, were removed to store the values as numbers. The amounts were then converted from Canadian dollars to US dollars using exchange rates corresponding to the year of each record. CAFC also uses \$0.00 for identity fraud, personal information and phishing when the loss cannot be determined. These values were therefore changed to NULL, while \$0.00 for other fraud types was retained as 0, as it represents no reported financial loss.

The dates were already in ISO format (YYYY-MM-DD), the same as in the mock data, so no conversion was required. Furthermore we checked the CAFC data for duplicate records and found no duplicate report numbers in the data. Finally, the data contained every column twice, once in English and once in French. To maintain consistent naming conventions in our database, only the English versions were used.

The FTC data required the schema to create two new tables. The FTC only gives totals per year and not separate reports, so the data does not fit in Person, Fraud_Incident or Report.

We added Report_Count for the number of reports per fraud type and year, and Loss_Statistic for the losses per contact method or age group. We also made Fraud_kind and Channel_kind case-insensitive, so "Phone call" and "Phone Call" are not stored twice.

The FTC data also needed to be transformed. Missing values are not marked: if someone did not fill in their age, the report is just left out of that table. There are no full dates, only years, which we stored as Report_year. One report can be counted in more than one subcategory, so the subcategories together are higher than the category total. Some names have a symbol behind them, like "Eye Care*", which we removed. Amounts like "\$502M" and percentages like "25%" were changed to numbers. Lastly, the FTC names were changed to the names the CAFC data already uses, for example "Text" became "Text message".

Example queries week 3

We re-ran the three example queries from week 3 using the real CAFC and FTC data and checked whether the results were as expected.

Query 1 (organisations ranked by number of reports) works without changes.

Query 2 (average loss by age bracket and channel) needed to be updated. The CAFC data mainly contains age ranges instead of exact ages. We therefore changed the query to use Age when available and Age_range otherwise, with "Unknown" for missing ages.

Query 3 (repeat victims) works correctly with the same logic as before. The query returns no rows for the real data because every person in the sample has only one incident. This is a meaningful result rather than an error. The query was also confirmed to work using the mock data, where repeat victims were found.

Query 4 (Canada vs. the United States) was added to compare the fraud types in the two real datasets. We added Query 4 so that we could use the real data and get an actual result from the query, since Query 3 does not return any results with the real data. It ranks the fraud type in each country and shows the difference between their shares.

Normalization realworld data

With the CAFC and FTC data the database is still normalized up to 3NF, no further adjustments had to be made.

Results queries

Overall, the queries produce meaningful results after the necessary updates. Query 1 correctly reflects how reports are stored, Query 2 provides useful information about the average loss by age and channel. Query 3 correctly shows that there are no repeat victims in the current sample. Query 4 also gives a useful comparison between Canada and the US, showing that Identity Fraud is the largest category in both, while some other fraud types have different shares between the countries. The results show that Identity Fraud is the most common fraud type in both countries, accounting for 62.5% of CAFC reports and 58.7% of FTC reports.

Evaluation of Query Results

Query 1: Expected. All 200 CAFC reports are linked to the Canadian Anti-Fraud Centre. The FTC data is aggregated, so no individual reports are linked to it and it shows 0.

Query 2: Partly expected. Phone calls causing high losses for ages 50-69 fits what I assumed. The very high averages for "Internet" (30-49) and "Unknown" were less expected, but they are based on few incidents, so a few large losses dominate. Several 0.00 averages mean those incidents had no loss.

Query 3: Expected. Each report in the CAFC sample belongs to a different person, so there are no repeat victims. The dataset cannot show repeat victimisation over time.

Query 4: Partly expected. Identity Fraud is the largest type in both countries (62.5% CAFC, 58.7% FTC, calculated within the fraud types present in both datasets). The biggest differences are Merchandise fraud (higher in the US) and Investment fraud (higher in Canada). Because the CAFC data is a sample, small differences should not be over-interpreted.

Results

===== Query 1 =====

	Organisation_name	Organisation_type	reports_handled
0	Canadian Anti-Fraud Centre	Law Enforcement	200
1	FTC Consumer Sentinel Network	Government	0

===== Query 2 =====

	age_bracket	Channel_kind	incident_count	avg_loss	loss_rank
0	30-49	Internet	9	59901.74	1
1	30-49	Internet-social network	7	8411.80	2
2	30-49	Direct call	5	2031.77	3
3	50-69	Direct call	8	54625.00	1
4	50-69	Other/unknown	2	17945.00	2
5	50-69	Internet	8	12023.92	3
6	50-69	Internet-social network	1	70.20	4
7	50-69	Email	3	0.00	5
8	70+	Internet	1	15702.00	1
9	70+	Internet-social network	3	9104.50	2
10	70+	Door to door/in person	1	1112.00	3
11	70+	Direct call	1	0.00	4
12	70+	Email	1	0.00	4
13	Under 30	Internet	4	968.45	1
14	Under 30	Email	3	333.33	2
15	Under 30	Internet-social network	3	282.91	3
16	Under 30	Direct call	2	0.00	4
17	Unknown	Email	7	106579.61	1
18	Unknown	Text message	3	73595.57	2
19	Unknown	Internet-social network	7	20911.00	3
20	Unknown	Door to door/in person	1	7938.00	4
21	Unknown	Internet	4	6227.50	5
22	Unknown	Direct call	5	6202.11	6
23	Unknown	Not Available	10	131.50	7

===== Query 3 =====

Empty DataFrame

Columns: [Person_ID, Occupation, Age, Age_range, incident_count, total_loss]

Index: []

===== Query 4 =====

	Fraud_kind	cafc_reports	cafc_share_pct	canada_rank	ftc_reports
0	Merchandise	15	12.5	2	383441
1	Investments	14	11.7	3	116071
2	Prize	1	0.8	7	97350
3	Identity Fraud	75	62.5	1	1135291
4	Romance	5	4.2	5	59490
5	Loan	2	1.7	6	23395
6	Job	7	5.8	4	104946
7	Fraudulent Cheque	1	0.8	7	13616

	ftc_share_pct	us_rank	share_difference_pct
0	19.8	2	-7.3
1	6.0	3	5.7

2	5.0	5	-4.2
3	58.7	1	3.8
4	3.1	6	1.1
5	1.2	7	0.5
6	5.4	4	0.4
7	0.7	8	0.1